

Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — RA-001

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cellular cytotoxicity; CDC, complement dependent cytotoxicity; CEX, cation exchange chromatography; CofA, certificate of analysis; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DoE, design of experiments; DS, drug substance; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; pCO₂, dissolved carbon dioxide partial pressure; RPN, risk priority number; SEC, size exclusion chromatography; SOP, standard operating procedure; UF/DF, ultrafiltration and diafiltration; XMuLV, xenotropic murine leukaemia virus.

1 Purpose and scope

This assessment decides which process parameters of the A-Mab drug substance process are characterized, and what kind of study each one receives. It is performed before any characterization study is executed, so it draws on prior knowledge, the platform history and the process description alone. Its output is a scope. For every parameter carried in the process description it records the quality attribute that the parameter could put at risk, an initial risk priority number, and the study type assigned to it.

The assessment does not classify parameters. Classification into critical, well controlled critical, key and general process parameters is an outcome of the studies, and it is recorded in the characterization reports PCR-003 through PCR-010. The assessment also defines no operating range and no design space, and it states only what must be studied and why.

The scope is the drug substance train described in PTP-001, which runs from the production bioreactor to ultrafiltration and diafiltration. That train has 8 unit operations and 37 process parameters, and it must control 10 drug substance quality attributes, all of which are carried into the assessment. Vial thaw and seed expansion are not included, and neither are the drug product operations. Analytical methods, raw materials and facility systems are qualified under their own procedures and fall outside an assessment of process parameters.

PTP-001 is the parent of this document (Table 1.1), and the scope defined here is consolidated across the campaign in PCMP-001 before it is instantiated step by step in the characterization plans PCP-003 through PCP-010. Each plan sets the ranges to be studied and the acceptance criteria for its own step. A plan that departs from the scope assigned here records the reason for the departure.

Table 1.1: Documents that frame this assessment.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Risk assessment methodology

Two decisions are taken in this document and they draw on different parts of the A-Mab method (CMC Biotech Working Group 2009). Parameters are ranked on the severity, occurrence and detection scales of the case study, which the case study applies after characterization and which are applied here prospectively, at a point in development where no characterization data exist. The study type assigned to each parameter follows the case study's risk ranking and filtering rule, and that rule is given in §2.4. Each parameter receives three scores. Severity comes from the quality attribute that the parameter could affect, occurrence describes how likely an effect on that attribute is thought to be, and detection describes whether the controls that exist today would find the failure, and how late. The product of the three is the risk priority number (RPN), and parameters are ranked on it.

2.1 Severity

Severity is a property of the quality attribute and not of the parameter, so a parameter is scored at the severity of the most severe attribute it could affect, and a parameter that could move high mannose sits at the top of the scale even where the movement expected is small. The scale runs from a deviation with no product quality consequence to one that would cause a batch to be rejected (Table 2.1). The priority band in the step tables of §4 follows the severity alone. A parameter linked to an attribute of severity 8 or above takes the High band, a parameter linked to a less severe attribute takes the Medium band, and a parameter that reaches no attribute takes the Low band.

Table 2.1: Severity scale for the effect of a parameter excursion.

Score	Band	Meaning
10	Very High	Definite impact to product quality; lot(s) rejected
8	High	Probable impact; significant supplemental testing / re-processing
6	Moderate	Potential impact; supplemental t=0 testing / minor re-processing
4	Slight	Unlikely impact; no supplemental testing / no re-processing
2	Low/None	No impact to product quality

2.2 Occurrence and detection before characterization

Occurrence and detection are scored by convention in this assessment, because neither can be measured before the studies are run. A parameter that is linked to a quality attribute is given the occurrence score for an effect that is credible but not yet quantified. A parameter that affects only process performance is given a lower score, since the platform history for such parameters is long and an excursion costs performance without putting an attribute at risk.

Table 2.2: Occurrence scale for a parameter excursion.

Score	Band	Meaning
10	Very High	Failure ~ once per 100 units
8	High	Failure ~ once per 1,000 units
6	Moderate	Failure ~ once per 2,000 units
4	Low	Failure ~ once per 5,000 units
2	Minimal	Failure ~ once per 10,000 units or lower

The scale in Table 2.2 is the case study scale and it counts how often an excursion happens. Occurrence carries a different meaning here. It stands for the likelihood that an excursion moves the attribute, which is the quantity the characterization studies are run to settle, so a parameter linked to a quality attribute is scored between the Moderate and High bands of the scale, and a parameter that affects only process performance is scored at the Low band.

Detection follows the control that exists today (in-process monitoring, in-process testing and release testing), and the scale is given in Table 2.3. A parameter that could reduce viral clearance receives the worst detection score, because clearance is not measured on a routine batch and its loss would not appear in release testing.

Table 2.3: Detection scale for the resulting failure.

Score	Band	Meaning
10	None	Not detected by in-process or CofA testing
8	Low	Not detected in-process; caught by CofA testing
6	Moderate	Detected downstream before CofA, not during the unit op
4	High	Detected in the next downstream operation
2	Very High	Immediately detected by in-process monitoring / inspection

The three conventions, and the number of parameters each one covers, are given in Table 2.4. The classes are mutually exclusive, so the counts in the last column sum to the 37 parameters in the assessment, of which viral safety accounts for 8.

Table 2.4: Pre-characterization occurrence and detection scores by parameter class.

Parameter class	Occurrence	Detection	Parameters
Impacts a viral-safety attribute	7	10	8
Impacts another quality attribute	7	8	13
Process performance only	4	6	16

2.3 Criticality and parameter risk are separate

Two assessments meet in this document and they are not the same thing. The criticality of a quality attribute is scored on impact and uncertainty, which is Tool #1 of the case study, and it produces a continuum of criticality rather than a division into critical and not critical (CMC Biotech Working Group 2009; International Council for Harmonisation 2023). Attribute criticality is a property of the molecule and of what the attribute does in the patient, whereas the risk of a process parameter is scored on severity, occurrence and detection and is a property of the process and of how well that process is currently understood. An attribute of very high criticality does not make every parameter that touches it a high risk parameter, and a parameter can score high because the process cannot yet detect the failure, even where the attribute itself is moderate.

2.4 The study type decision

Two things decide the study type, and the initial RPN is not one of them (Table 2.5). The first is whether an excursion in the parameter could reach a quality attribute. The second is whether the effect of the parameter is expected to depend on the other parameters at the same step. A parameter that meets both is carried into the designed experiment for its unit operation, so that its effect is estimated alongside the other parameters at that step and alongside the interactions between them. A parameter that is linked to an attribute but whose effect is expected to be independent is studied on its own, and the reason is stated with the assignment. A parameter that affects only process performance is also studied on its own, unless it is a principal driver of a response that the designed experiment at its step already measures, in which case it is carried as a factor of that design. The rule allows a parameter to be left unstudied as well, and that outcome requires prior knowledge that the parameter reaches no attribute and that the range given in the process description is already justified. No parameter in this assessment met both requirements, and the count is given in the table.

Table 2.5: Study-type decision rule and the number of parameters reaching each outcome.

Study type	Basis for the decision	Parameters
multivariate DoE	Effect expected to depend on the other parameters at the step	22
univariate (justified)	Impacts a quality attribute, studied outside the designed experiment	1
univariate	Process performance only, no expected interaction at the step	14
no study	No attribute risk and a range already justified by prior knowledge	0

2.5 What this assessment does not produce

Three outputs lie outside this assessment by design: it does not estimate the size of an effect, it does not resolve the interactions between parameters, and it sets no proven acceptable range. It also produces no residual risk number, and a residual assessment needs the characterization results and the control strategy that follows from them, so it is performed after the studies report and is consolidated in PCMR-001.

3 Quality attributes at risk

The severity axis of this assessment comes from the drug substance quality attribute register, which is reproduced in Table 3.1. The register carries 10 attributes, of which 6 carry a high or very high criticality designation. Criticality and severity appear as separate columns because they answer separate questions: criticality states how much the attribute matters to the patient and how certain that judgement is, and severity states what would happen to a batch if the attribute moved outside its acceptance range.

Table 3.1: Drug substance quality attributes, their acceptance criteria, criticality and severity.

Quality attribute	Cate- gory	Acceptance	Criti- cality	Tool #1	Sever- ity	Set by
Afucosylation	glycosy- lation	2-13 % of glycans	H	60	8	Production Bioreactor
Galactosylation (total %Gal)	glycosy- lation	10-40 % (G1+G2 equiv.)	H	48	8	Production Bioreactor
High Mannose	glycosy- lation	3-10 % of glycans	VH	80	10	Production Bioreactor
Aggregates (HMW)	size variant	0-5 % HMW (SEC)	H	60	8	Production Bioreactor
Acidic charge variants (deamidation)	charge variant	18-40 % (CEX/icIEF)	VL	4	4	Production Bioreactor
Host Cell Protein (HCP)	process impurity	0-100 ng/mg	M-H	36	6	Production Bioreactor
Residual DNA	process impurity	0-0.001 ng/dose	VL	6	2	Production Bioreactor
Leached Protein A	process impurity	0-5 ppm	L-M	16	4	Protein A Chromatogra- phy
Viral clearance — XMuLV (enveloped)	viral safety	16.7-30 log10 (cumulative)	VH	20	10	Low-pH Viral Inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6-20 log10 (cumulative)	VH	20	10	Anion Exchange Chromatogra- phy

The top of the severity scale holds 3 attributes. High mannose carries the highest Tool #1 score in the register and is formed in the culture. The two viral clearance attributes are expressed as a cumulative log reduction across the whole process, so their acceptance criteria are a floor and not a window. A batch that fails either of them cannot be brought back by reprocessing, and that is what puts both at the top of the scale.

Afucosylation is scored high because the mechanism of action of A-Mab is antibody dependent cellular cytotoxicity (ADCC) and afucosylation moves that activity. Galactosylation is scored high on a different activity, since the case study reports a correlation between galactose content and complement dependent cytotoxicity (CDC) across the whole galactosylation range (CMC Biotech Working Group 2009). Both are formed in the production bioreactor. Prior knowledge from the platform indicates that neither is modified by the purification train, so the culture parameters carry the whole of the risk for them and no downstream step can correct an excursion.

The process impurities behave differently, since host cell protein (HCP), residual DNA and leached Protein A are reduced at more than one step and no single unit operation carries the whole risk for any of them. The same holds for the two viral clearance attributes, and the two are built differently. The enveloped virus claim takes independent contributions from the low pH hold, the anion exchange step and the virus filter. The parvovirus claim rests on the anion exchange step and the virus filter alone, because a low pH hold does not inactivate a virus without an envelope. Cross step credit is part of the assessment for these attributes, so a parameter is scored on the attribute it puts at risk and the report for each contributing step states how much of the attribute that step delivers.

The production bioreactor sets 7 of the 10 attributes, and the remaining 3 are set at the capture step, the low pH hold and the anion exchange step (Table 3.1). That distribution is the main reason the bioreactor carries the largest parameter set in this assessment.

4 Parameter risk ranking by unit operation

The parameter level outcome is given step by step below, and Table 4.1 gives its shape: the number of parameters each step carries, how many of them are linked to a quality attribute, the split between multivariate and univariate study, and the highest initial RPN reached at the step.

Table 4.1: Parameters carried, study split and highest initial RPN, by unit operation.

Step	Unit operation	Parameters	CQA-linked	Multivariate	Univariate	Highest init. RPN
3	Production Bioreactor	9	5	5	4	560
4	Harvest and Clarification	3	0	0	3	96
5	Protein A Chromatography	6	2	4	2	336
6	Low-pH Viral Inactivation	4	3	3	1	700
7	Cation Exchange Chromatography	5	4	4	1	448
8	Anion Exchange Chromatography	5	5	4	1	700
9	Small-Virus Retentive Filtration	2	2	2	0	700
10	Ultrafiltration / Diafiltration	3	0	0	3	96

The highest initial RPN in the assessment is reached at 3 steps, and all of them are viral steps. That number is shared by 8 parameters, and it is the product of the top severity, the pre-characterization occurrence convention and the worst detection score. 2 steps, harvest and UF/DF, carry no parameter linked to a quality attribute at all, and both are assessed on process performance alone. The bioreactor sits below the viral steps on initial RPN and above every other step on parameter count.

Each step below carries two tables. The first ranks the parameters of the step and gives the three scores behind the ranking. The second records the prospective failure mode and its effect. Where a failure mode quotes a range or a limit, that value is taken from the process description, and none of it is an outcome of a characterization study.

4.1 Production bioreactor (Step 3)

The bioreactor carries more parameters than any other step, and it is where the glycan, charge variant, aggregate and impurity attributes are formed. It carries 9 parameters, of which 5 are linked to a quality attribute (Table 4.2). Culture pH reaches the highest initial RPN at the step because it is the only bioreactor parameter that could move high mannose, which is the one attribute the step forms that sits at the top of the severity scale. Culture temperature, dissolved carbon dioxide (pCO₂), osmolality and culture duration are scored one severity band lower, on the attributes each of them reaches.

Table 4.2: Production bioreactor parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Pri- or- ity	Study type
Culture pH	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)	10	7	8	560	High	multi-variate DoE
Culture temperature	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	7	8	448	High	multi-variate DoE
Dissolved CO ₂ (pCO ₂)	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	7	8	448	High	multi-variate DoE
Osmolality	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	7	8	448	High	multi-variate DoE
Culture duration	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA	8	7	8	448	High	multi-variate DoE
Dissolved oxygen	process performance	4	4	6	96	Low	uni-variate
Initial viable cell conc.	process performance	4	4	6	96	Low	uni-variate
Basal medium concentration	process performance	4	4	6	96	Low	uni-variate

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Pri- or- ity	Study type
Nutrient feed-1 volume	process performance	4	4	6	96	Low	uni-variate

The prospective failure modes are given in Table 4.3, and they act by two different mechanisms. A pH or temperature excursion acts on the glycosylation machinery directly, and the effect is expected to be continuous across the range. Culture duration acts indirectly, because an extended culture is harvested at lower viability, which raises host cell protein and DNA in the harvest and can raise aggregate as well. Separating a direct effect from an indirect one, and separating both from the interactions between the parameters carried into the design, is what the designed experiment at this step is for.

Table 4.3: Production bioreactor prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Culture pH	pH excursion outside the range proposed for characterization (6.6-7.1)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile
Culture temperature	Temperature excursion outside 34-36 C	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH
Dissolved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Strong effect on galactosylation and acidic variants
Osmolality	Osmolality outside 360-440 mOsm	Affects galactosylation and acidic variants
Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines
Dissolved oxygen	DO outside 30-70%	Impacts peak VCD, titer and process consistency (no significant CQA impact)
Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	Drives peak/integral VCD and titer; no product-quality impact
Basal medium concentration	Basal medium concentration outside 0.8-1.6X	Statistically significant but shallow effect on afucosylation; drives titer
Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	Slight effect on galactosylation; critical for titer

Dissolved oxygen, initial viable cell concentration, basal medium concentration and nutrient feed volume are scored on process performance. They drive titre and culture consistency (peak cell density, integral of viable cells, final titre), and no product quality effect is expected for them over the ranges the process uses. Basal medium concentration is a partial exception, because prior product knowledge records a statistically significant but shallow effect of medium concentration on afucosylation, and that effect is too small to move the attribute within its acceptance range, so the parameter is carried on process performance and studied on its own.

4.2 Harvest and clarification (Step 4)

Harvest forms no quality attribute and is credited with no clearance, and the 3 parameters it carries govern clarification efficiency and filter capacity, each of them scoring 96, the lowest initial RPN in the assessment (Table 4.4). The recorded failure modes are losses of yield or of filter capacity (Table 4.5). Feed quality to the capture step is bounded by the post clarification turbidity range in the process description, which runs from 1 to 20 NTU with a normal operating range of 1 to 10 NTU. Turbidity is an in-process limit and not an operator set point, and it is carried here because it bounds the material the capture step receives.

Table 4.4: Harvest and clarification parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Prior-ity	Study type
Centrifugation (rcf)	process performance	4	4	6	96	Low	univariate
Depth filter load	process performance	4	4	6	96	Low	univariate
Post-clarification turbidity	process performance	4	4	6	96	Low	univariate

Table 4.5: Harvest and clarification prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	Affects clarification/yield (no CQA impact)
Depth filter load	Depth filter load outside 80-160 L/m2	Affects filter capacity/turbidity (no CQA impact)
Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)

4.3 Protein A chromatography (Step 5)

Protein A capture sets leached Protein A and removes the bulk of the host cell protein and DNA load that enters the purification train. The largest further reductions come later, at cation exchange for host cell protein and at anion exchange for DNA. The step carries 6 parameters, of which 2 are linked to a quality attribute, and both of those are linked to host cell protein (Table 4.6). Protein load and elution buffer pH reach the same initial RPN, and the failure modes recorded for them share a mechanism. Both increase the amount of host cell protein that leaves the step with the product, and the assessment expects the two to interact (Table 4.7).

Table 4.6: Protein A chromatography parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Prior-ity	Study type
Protein load	Host Cell Protein (HCP)	6	7	8	336	Medium	multivariate DoE
Elution buffer pH	Host Cell Protein (HCP)	6	7	8	336	Medium	multivariate DoE
Load flow rate	process performance	4	4	6	96	Low	multivariate DoE
End of pool collect	process performance	4	4	6	96	Low	multivariate DoE
Operating temperature	process performance	4	4	6	96	Low	univariate
Bed height	process performance	4	4	6	96	Low	univariate

Load flow rate and end of pool collect carry no attribute risk. Both are taken into the designed experiment even so, because step yield is measured on the same runs and these are the two parameters that move it. Operating temperature and bed height carry no specific failure mode, and the generic entry recorded against them states the position the assessment takes. No significant product quality impact is expected from either of them, even for an excursion beyond the range proposed for characterization, and a univariate study is enough to support that range.

Table 4.7: Protein A chromatography prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Protein load	Protein load above the range proposed for characterization (>50 g/L resin)	Increases pool HCP (interaction with elution pH)
Elution buffer pH	Elution pH below the range proposed for characterization (<3.2)	Increases pool HCP; also sets up viral inactivation feed
Load flow rate	Load flow rate high with high load	Reduces product yield (no CQA impact)

Parameter	Potential failure mode	Potential effect
End of pool collect	End-of-pool collection outside 2.0-3.2 CV	Affects product yield (no CQA impact)
Operating temperature	Parameter outside the range proposed for characterization	No significant product-quality impact expected
Bed height	Parameter outside the range proposed for characterization	No significant product-quality impact expected

4.4 Low-pH viral inactivation (Step 6)

The low pH hold reaches the highest initial RPN in the assessment, and it does so on 3 of its 4 parameters (Table 4.8). The step sets the cumulative clearance of enveloped virus (XMuLV), which is an attribute at the top of the severity scale that no in-process test can measure, and inactivation pH, hold time and temperature are all scored there.

Table 4.8: Low-pH viral inactivation parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Priority	Study type
Inactivation pH	Viral clearance — XMuLV (enveloped), Aggregates (HMW)	10	7	10	700	High	multi-variate DoE
Hold time	Aggregates (HMW), Viral clearance — XMuLV (enveloped)	10	7	10	700	High	multi-variate DoE
Temperature	Viral clearance — XMuLV (enveloped)	10	7	10	700	High	multi-variate DoE
A-Mab concentration	process performance	4	4	6	96	Low	univariate

Inactivation pH is the sharpest case in the assessment, because it carries a failure mode in both directions. Too high a pH leaves the inactivation incomplete, and too low a pH risks aggregation and precipitation of the product (Table 4.9). The two failure modes act on different attributes and bound the acceptable range from both sides, which is why the parameter is taken into a designed experiment together with hold time and temperature.

Table 4.9: Low-pH viral inactivation prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Loss of XMuLV inactivation assurance or increased aggregate
Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Incomplete inactivation or increased aggregate
Temperature	Temperature < 15 C	Lower inactivation rate at short time / higher pH
A-Mab concentration	Concentration > 35 g/L	No significant effect on inactivation, aggregation or charge variants

A-Mab concentration in the hold is scored on process performance. No effect on inactivation kinetics, aggregation or charge variants is expected over the concentration range the step sees, and the parameter is assigned a univariate study.

4.5 Cation exchange chromatography (Step 7)

Cation exchange is the principal aggregate polishing step and a major clearance step for host cell protein, DNA and leached Protein A. It carries 5 parameters, of which 4 are linked to a quality attribute (Table 4.10). Protein load, elution buffer pH and elution stop collect are scored on aggregate and reach the highest initial RPN at the step, while load and wash conductivity is scored on host cell protein, one severity band lower.

Table 4.10: Cation exchange chromatography parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Priority	Study type
Protein load	Aggregates (HMW), Host Cell Protein (HCP)	8	7	8	448	High	multivariate DoE
Load/Wash conductivity	Host Cell Protein (HCP)	6	7	8	336	Medium	multivariate DoE
Elution buffer pH	Aggregates (HMW)	8	7	8	448	High	multivariate DoE
Elution stop collect	Aggregates (HMW)	8	7	8	448	High	multivariate DoE
Elution flow rate	process performance	4	4	6	96	Low	univariate

The failure modes at this step are all reductions in separation quality (Table 4.11), because a high load and an extended stop collect both bring more of the aggregate peak into the pool, and elution pH moves where the aggregate elutes relative to the monomer. The assessment expects load, elution pH and wash conductivity to interact, which is why 4 parameters go into one designed experiment. Elution flow rate affects peak shape and yield and is studied on its own.

Table 4.11: Cation exchange chromatography prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Protein load	Protein load above the range proposed for characterization (>40 g/L resin)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)
Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Reduced HCP clearance
Elution buffer pH	Elution pH outside 5.8-6.2	Alters aggregate distribution and yield
Elution stop collect	Extended elution stop-collect (>1.5 OD)	Minor increase in pool aggregate
Elution flow rate	Elution flow rate outside 100-300 cm/hr	Affects peak shape and yield (minor aggregate)

4.6 Anion exchange chromatography (Step 8)

Anion exchange and small virus retentive filtration are the two steps at which every parameter carried is linked to a quality attribute, and anion exchange is the larger of the two (Table 4.12). It is the last chromatographic step in the train, it sets the cumulative clearance of parvovirus (MVM), and it also contributes to enveloped virus clearance and to host cell protein removal. Load pH, load conductivity and operating flow rate are scored on viral clearance and reach the highest initial RPN in the assessment, while equilibration and wash conductivity and protein load are scored on host cell protein.

Table 4.12: Anion exchange chromatography parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Prior-ity	Study type
Load pH	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	7	10	700	High	multi-variate DoE

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Pri- or- ity	Study type
Equil/Wash-1 conductivity	Host Cell Protein (HCP)	6	7	8	336	Medium	multivariate DoE
Load conductivity	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	7	10	700	High	multivariate DoE
Protein load	Host Cell Protein (HCP)	6	7	8	336	Medium	multivariate DoE
Operating flow rate	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	7	10	700	High	univariate (justified)

Operating flow rate is the one parameter in the assessment that reaches the highest initial RPN and is still not assigned to a designed experiment. The reason is mechanistic. Flow rate is expected to act through residence time, and not through the ionic conditions that govern how impurities and virus bind to the resin, so its effect should not interact with load pH or with conductivity. The recorded failure mode is a loss of clearance margin at high flow (Table 4.13). The parameter is assigned a justified univariate study, and PCP-008 records that justification together with the range it studies. If the univariate result does not support the expectation, the parameter returns to the multivariate scope.

Table 4.13: Anion exchange chromatography prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Load pH	Load pH below the range proposed for characterization (<7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance
Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal
Load conductivity	Load conductivity above the range proposed for characterization	Reduced viral clearance (interacts with pH)
Protein load	Protein load above the range proposed for characterization (>300 g/L resin)	Reduced HCP removal capacity (viral clearance robust)
Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin

4.7 Small-virus retentive filtration (Step 9)

Virus filtration carries 2 parameters and both are linked to viral clearance (Table 4.14). The step provides the dedicated size based removal of small virus and is the principal contributor to the parvovirus claim, so both parameters reach the highest initial RPN in the assessment. Filtration volume is the parameter the parvovirus claim is most sensitive to. Prior knowledge for this filter type indicates that the log reduction for parvovirus falls as the volumetric load rises, and the failure mode recorded against it is breakthrough above the control limit (Table 4.15). Filtration pressure is scored with it, because pressure and load act together on the filter.

Table 4.14: Small-virus retentive filtration parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Pri- ority	Study type
Filtration volume (load)	Viral clearance — MVM (parvovirus)	10	7	10	700	High	multi-variate DoE
Filtration pressure	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)	10	7	10	700	High	multi-variate DoE

Table 4.15: Small-virus retentive filtration prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Filtration volume (load)	Volumetric load above 105 L/m2	Reduced MVM log-reduction (parvovirus breakthrough risk)
Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load

4.8 Ultrafiltration and diafiltration (Step 10)

Ultrafiltration and diafiltration concentrates the product and exchanges it into the formulation buffer, and the step forms no quality attribute and is credited with no clearance, so all 3 of its parameters score 96 (Table 4.16). The recorded failure modes are incomplete buffer exchange, a loss of flux and a final concentration outside the drug substance specification (Table 4.17), and all three parameters are assessed univariately.

Table 4.16: Ultrafiltration and diafiltration parameter risk ranking.

Parameter	CQA(s) at risk	Sev.	Occ.	Det.	Init. RPN	Priority	Study type
Number of diavolumes	process performance	4	4	6	96	Low	univariate
Transmembrane pressure	process performance	4	4	6	96	Low	univariate
Final DS concentration	process performance	4	4	6	96	Low	univariate

Table 4.17: Ultrafiltration and diafiltration prospective failure modes and effects.

Parameter	Potential failure mode	Potential effect
Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)
Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)
Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)

5 Characterization study assignment

The assessment assigns a study to every parameter it carries: 22 parameters go into a multivariate designed experiment, 1 is assigned a justified univariate study, and 14 are assessed univariately on process performance. No parameter was filtered out as too low a risk to study. The parameter list contains only settings and in-process limits that the operator controls, and each of those is given at least the univariate work needed to justify its range.

Table 5.1: Characterization scope handed to each plan, by unit operation.

Unit				
Step	operation	Multivariate DoE	Univariate	Documents
3	Production Bioreactor	Culture pH, Culture temperature, Dissolved CO ₂ (pCO ₂), Osmolality, Culture duration	Dissolved oxygen, Initial viable cell conc., Basal medium concentration, Nutrient feed-1 volume	PCP-003 / PCR-003
4	Harvest and Clarification	none	Centrifugation (rcf), Depth filter load, Post-clarification turbidity	PCP-004 / PCR-004
5	Protein A Chromatography	Protein load, Elution buffer pH, Load flow rate, End of pool collect	Operating temperature, Bed height	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	Inactivation pH, Hold time, Temperature	A-Mab concentration	PCP-006 / PCR-006
7	Cation Exchange Chromatography	Protein load, Load/Wash conductivity, Elution buffer pH, Elution stop collect	Elution flow rate	PCP-007 / PCR-007
8	Anion Exchange Chromatography	Load pH, Equil/Wash-1 conductivity, Load conductivity, Protein load	Operating flow rate (justified)	PCP-008 / PCR-008

Unit				Docu- ments
Step	operation	Multivariate DoE	Univariate	
9	Small- Virus Retentive Filtration	Filtration volume (load), Filtration pressure	none	PCP- 009 / PCR- 009
10	Ultrafiltra- tion / Diafiltra- tion	none	Number of diavolumes, Transmembrane pressure, Final DS concentration	PCP- 010 / PCR- 010

The assignment by step is given in Table 5.1. A designed experiment is carried at 6 of the 8 steps. Harvest and UF/DF do not, because no parameter at either is linked to a quality attribute, and a multivariate design would resolve interactions with no quality consequence.

The multivariate group is the larger one and it is defined by attribute risk, with two additions at the capture step. A parameter enters the designed experiment for its step when an excursion could move an attribute and when its effect is expected to depend on the other parameters at that step. Load, pH and conductivity at the chromatography steps are the clearest example of that general case. Each of them changes the chemical environment in which the product and the impurities bind, so their effects cannot be estimated one by one and then added together. The two additions are load flow rate and end of pool collect at the capture step, which reach no attribute and are carried because step yield is measured on the same runs.

The univariate group divides into two, and 14 parameters affect process performance alone, for which a univariate study establishes the range over which the performance measures (titre, yield, filter capacity) stay acceptable. The anion exchange operating flow rate is the only parameter linked to a quality attribute that is assigned a univariate study, and its justification is mechanistic and is given in the anion exchange section above.

This assignment is prospective, and it is bounded in three ways. It rests on prior knowledge of the platform and of the three antibodies made on it before A-Mab, so it assumes that platform behaviour transfers. It fixes a study type but not a range. Each plan sets the ranges to be studied, and a range wider than the routine control range is what makes the study informative (CMC Biotech Working Group 2009). The assignment is also revisited when the studies report. A parameter that turns out to carry no effect keeps its place in the knowledge space as evidence of robustness, and a parameter that interacts more strongly than expected can be returned to a multivariate design in a later study.

6 Assumptions and limitations

The assessment assumes that the platform transfers, since A-Mab is made on the humanized IgG1 platform used for the three earlier products (X-Mab, Y-Mab and Z-Mab), and the failure modes recorded here are the platform's failure modes. Where the mechanism at a step is the platform mechanism, that assumption is strong. Where A-Mab differs from the earlier products, it is weaker, and the characterization studies are the check on it.

The parameter list is assumed to be complete. It holds the operating parameters named in the process description, and it does not hold material attributes, resin and membrane life cycle, or equipment design, all of which are controlled by other means. Leached Protein A is the clearest case. The register assigns that attribute to the capture step, and no operating parameter in the list is linked to it, because the amount that leaches is governed by the resin and by its sanitization history under SOP-2008. An assessment of operating parameters cannot cover an attribute of that kind, and the capture study measures it as a step response instead (PCP-005).

Occurrence and detection are conventions here and not measurements, and they are applied uniformly to whole classes of parameter, so the risk priority number ranks parameters against each other and does not measure risk on an absolute scale. Comparison of the number between two parameters is meaningful. Comparison of the number against an external threshold is not.

Severity is taken from the most severe attribute a parameter can affect, and nothing in the score rewards a parameter for affecting several attributes at once. Culture pH and culture duration show the limit of that convention. Each of them is linked to 5 attributes, and both still score below the viral parameters, which are linked to one or two. The ranking should be read with that in mind. The study assignment does not depend on it, because both parameters enter a designed experiment on either reading.

The assessment cannot know effect sizes or interactions. It can only say where they are plausible. Every quantitative statement about the effect of a parameter on an attribute belongs to the characterization reports, and nothing in this document should be read as one.

7 Outputs and downstream use

This assessment produces one output: the study type assigned to each process parameter, with the attribute risk that justifies it. PCMP-001 consolidates the assignment across the campaign and fixes the common scale down and statistical approach. Each plan from PCP-003 to PCP-010 takes the share belonging to its own step, sets the ranges and states the acceptance criteria, and each report from PCR-003 to PCR-010 records the outcome and classifies the parameters of that step. PCMR-001 consolidates the classifications and the resulting control strategy, and the whole package is listed in Table 7.1.

Table 7.1: The A-Mab drug substance document package.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)

Document ID	Document class	Subject
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

The assessment is re-executed when the process description changes, when a parameter is added to a step, or when a characterization study returns a result that the assessment did not anticipate. A re-execution is recorded as a new version of this document and is referenced by the plans it affects.

8 Approvals

This assessment is approved by the functions listed in the approval record at the front of the document (Table [1](#)). Approval by Quality Assurance releases the scope defined here for use in PCMP-001 and in the characterization plans PCP-003 through PCP-010.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.